FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Main)-2021

TEST DATE: 28-02-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. B

Sol. $\vec{v} = \alpha\hat{i} + \beta t\hat{j}$

$\vec{a} = \beta\hat{j}$

Speed $v = \sqrt{\alpha^2 + \beta^2 t^2}$

Tangential acceleration $a_t = \dfrac{dv}{dt}$

$a_t = \dfrac{d}{dt}\left(\sqrt{\alpha^2 + \beta^2 t^2}\right)$

At $t = \dfrac{\alpha\sqrt{3}}{\beta}$, $a_t = \dfrac{\sqrt{3}\beta}{2}$

Normal acceleration; $a_n = \sqrt{a^2 - a_t^2}$

$\therefore \quad a_n = \dfrac{\beta}{2}$

2. C

Sol. $U = \dfrac{1}{2}Kx^2 = 10J$

$W = \dfrac{1}{2}K\left[(2x)^2 - x^2\right] = 30\,J$

3. B

Sol. Net mass $M=\left[\frac{4}{3}\pi\left\{(2R)^3-R^3\right\}\right]\rho$

ρ = density

$\therefore \quad \rho=\frac{3M}{28\pi R^3}$

Now; $V=V_{2R_{\text{at centre}}}-V_{R_{\text{at centre}}}$

$V_{2R}=-\frac{GM_1}{2R}$; where $M_1=\frac{8}{7}M$

$V_R=-\frac{GM_2}{2R}$; where $M_2=\frac{M}{7}$

$\therefore \quad V=-\frac{G}{2R}\left(\frac{8M}{7}\right)-\left(-\frac{G}{2R}\left(\frac{M}{7}\right)\right)$

$\Rightarrow \quad V=-\frac{9GM}{14R}$

4. D

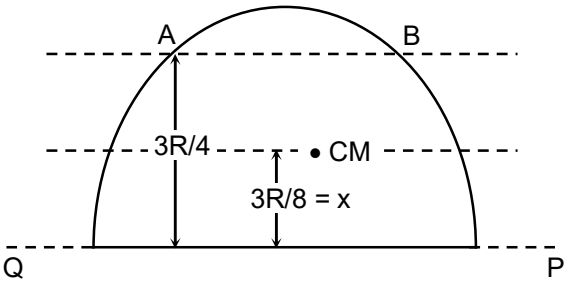
Sol. $I_{PQ}=I_{CM}+Mx^2$

$I_{AB}=I_{CM}+Mx^2$

$\therefore \quad I_{AB}=I_{PQ}$

$\Rightarrow \quad I_{AB}=\frac{2}{5}MR^2=MK^2$

$\therefore \quad K=\sqrt{\frac{2}{5}}R$



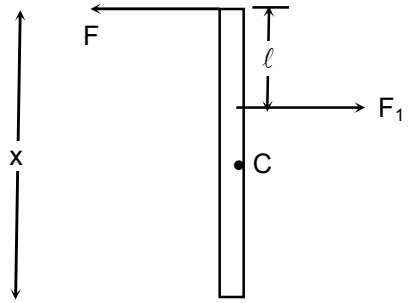
5. B

Sol. $F_1-F=ma$ …(1)

$F\left(\frac{x}{2}\right)=F_1\left(\frac{x}{2}-\ell\right)$ …(2)

Solving (1) & (2)

$x=2\ell\left(1+\frac{F}{ma}\right)$



6. C

Sol. $\frac{hc}{\lambda}-\phi=e(5V_0)$ …(1)

$\frac{hc}{3\lambda}-\phi=e(V_0)$ …(2)

Solving (1) & (2), we get

$\phi=\frac{hc}{6\lambda}$

7. A

Sol. $\frac{dN}{dt} = t^2 - \lambda N$

For $\frac{dN}{dt}$ to be minimum

$\frac{d^2N}{dt^2} = 2t - \lambda\frac{dN}{dt} = 0$

$\Rightarrow 2t - \lambda\left(t^2 - \lambda N\right) = 0$

At $t = t_0$

$N = \frac{\lambda t_0^2 - 2t_0}{\lambda^2}$

8. A

Sol. Frequency heard by 1st train due to second train

$\nu_1 = \left(\frac{300+20}{300-30}\right)\nu_0 = \frac{320}{270}\nu_0$

Frequency heard by 2nd train due to 1st train

$\nu_2 = \left(\frac{300+30}{300-20}\right)\nu_0 = \frac{330}{280}\nu_0$

$\therefore \quad \frac{\nu_1}{\nu_2} = \frac{320/270}{330/280} = \frac{896}{891}$

9. D

Sol. From $\rho = x$

Also, $\mu = \rho A = xA$

$V = \sqrt{\frac{T}{xA}}$

$\Rightarrow \frac{dx}{dt} = \sqrt{\frac{T}{xA}}$

$\Rightarrow \int_0^L \sqrt{x}\,dx = \sqrt{\frac{T}{A}}\int_0^t dt$

$\Rightarrow t = \frac{2}{3}\sqrt{\frac{A}{T}}L^{3/2}$

10. C

Sol. $\frac{Q_{in}}{W} = \frac{1}{\frac{T_2}{T_1} - 1} = \frac{T_1}{T_2 - T_1} = \frac{270}{30} = 9$

$W = \frac{1260}{9}\text{kJ/min} = 2.33\,\text{kJ/sec}$

11. D

Sol. We knot that mean collision time $\tau \propto \frac{V}{\sqrt{T}}$

$$TV^{\gamma-1} = c \Rightarrow T = \frac{c}{V^{\gamma-1}}$$

$$\therefore \ \tau \propto V^{1+\frac{\gamma-1}{2}}$$

$$\Rightarrow \ \tau \propto V^{\frac{\gamma+1}{2}}$$

$$\therefore \ \frac{\tau_2}{\tau_1} = \left(\frac{4V}{V}\right)^{\frac{\gamma+1}{2}} = (2)^{\gamma+1}$$

12. C

Sol. For r < R

$$\phi = \frac{1}{\varepsilon_0}\frac{Q}{R^3}r^3$$

For r > R

$$\phi = \frac{Q}{\varepsilon_0}$$

13. B

Sol. At t = 0, $I_1 = \frac{10}{5} = 2A$

After long time $I_2 = \frac{10}{1} = 10A$

$$\therefore \quad \frac{I_1}{I_2} = 0.2$$

14. A

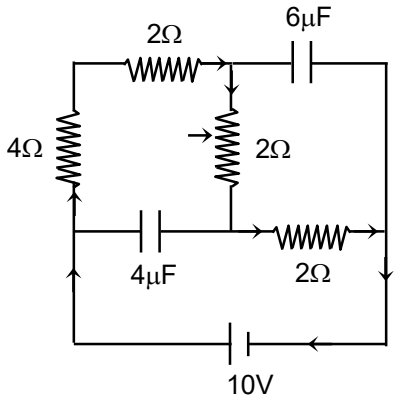
Sol. $$I_{rms} = \frac{10/\sqrt{2}}{\sqrt{5^2 + (7-2)^2}} = \frac{10/\sqrt{2}}{5\sqrt{2}} = 1A$$

$\therefore$ Reading of voltmeter
$= I_{rms}(X_L - X_C)$
$= (1)\ (7–2) = 5V$

15. B

Sol. $$I = \frac{10}{4+2+2+2} = 1A$$

$\therefore$ PD @ 6μF = 4V



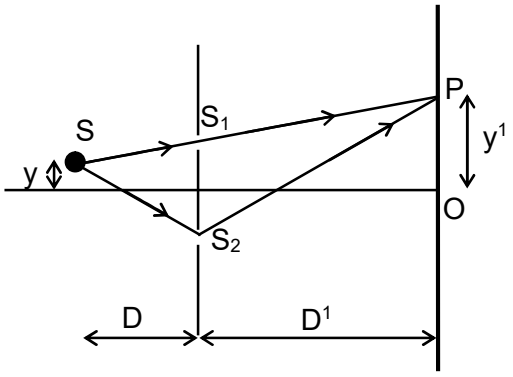
16. D

Sol. Surface energy $U = 2(4\pi r^2)\sigma$

Power required is $P = \dfrac{dU}{dt} = 16\pi r\sigma\alpha$

17. B

Sol. $y' = \dfrac{d}{2}$ for a point exactly in front S_1



So, $\Delta x = \dfrac{yd}{D} + \dfrac{\frac{d}{2}d}{D'}$

For maximum intensity $\Delta x = n\lambda$

$\dfrac{yd}{D} + \dfrac{\frac{d^2}{2}}{D'} = n\lambda$

$\Rightarrow 0.5\sin\pi t + \dfrac{(1)^2}{4} = n \times 500 \times 10^{-6} \times 10^3$

$\Rightarrow 0.5 \sin \pi t + 0.25 = 0.5n$

$\sin(\pi t) = \dfrac{0.5n - 0.25}{0.5}$

For minimum 't' we have n = 1

$\sin(\pi t) = \dfrac{0.5 - 0.25}{0.5} = 0.5$

$\pi t = \dfrac{\pi}{6}$

$t = \dfrac{1}{6}s$

18. A

Sol. $\vec{B} = 2\left\{\dfrac{\mu_0 I}{4\pi a}\left(\sin 60° + \sin 90°\right)\hat{k}\right\}$

$\therefore \vec{B} = \dfrac{\mu_0 I}{4\pi a}\left(2 + \sqrt{3}\right)\hat{k}$

19. C

Sol. $A = A_0 e^{-\frac{5t}{6}}$

$\Rightarrow \dfrac{A_0}{2} = A_0 e^{-\frac{5t}{6}}$

$\Rightarrow e^{-\frac{5t}{6}} = 2$

$\therefore t = \dfrac{6}{5}\ln 2$

20. C

Sol. λ_0 will change.

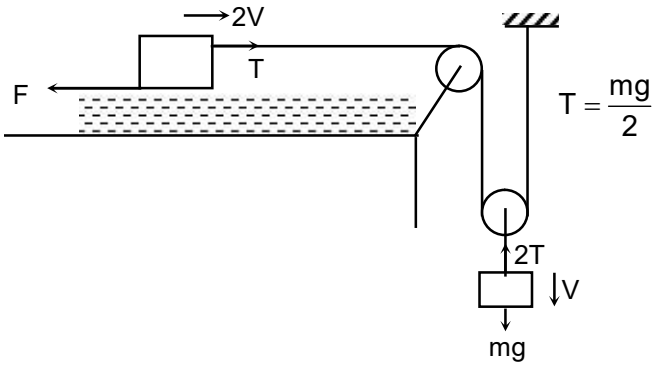
SECTION – B

21. 00005.00

Sol. For Block

$$\frac{mg}{2} = \eta A \frac{2V}{d}$$

$$\Rightarrow \eta = \frac{mgd}{4AV} = 5 \times 10^{-3} \text{ P}_a\text{-sec}$$



22. 00002.00

Sol. $a = \frac{T\sin\theta}{2m}$ ...(i)

$T\cos\theta = mg$ …(ii)

Restoring torque

$$\tau = -\left(mg\ell\sin\theta + \frac{mg}{2}\ell\sin\theta\right)$$

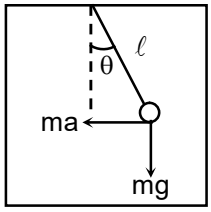
$$\Rightarrow m\ell^2\alpha = -\frac{3mg\ell}{2}\theta$$

$$\Rightarrow \alpha = -\left(\frac{3g}{2\ell}\right)\theta \quad \therefore \omega = \sqrt{\frac{3g}{2\ell}}$$

$$T = 2\pi\sqrt{\frac{2\ell}{3g}}$$

$$\therefore \quad K = \frac{2}{3} \Rightarrow 3K = 2$$

$\longrightarrow a = \frac{T\sin\theta}{2m}$



23. 00003.00

Sol. $-1.51\text{eV} = \frac{-13.6\,\text{eV}}{n^2}$

$$n^2 = \frac{13.6}{1.51} = 9$$

$$\therefore L = \frac{nh}{2\pi} = \frac{9h}{2\pi}$$

$\Rightarrow x = 3$

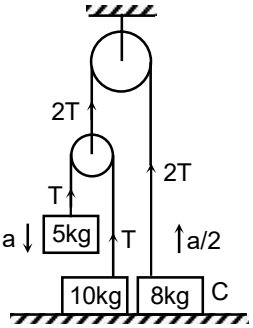
24. 00000.70

Sol. $5g - T = 5a$ …(1)

$2T - 8g = 8\dfrac{a}{2}$ …(2)

Solving (1) & (2)

$\dfrac{a}{2} = 0.7\ m/s^2$



25. 00003.00

Sol. No. of spectral lines = $^3C_2 = 3$

26. 00031.25

Sol. Energy released per fission = 200 MeV = $200 \times 1.6 \times 10^{-13}$J.

The rate at which energy is released = 1 Kw = 10^3 J/s.

$\therefore$ No. of fissions per second = $\dfrac{10^3}{200 \times 1.6 \times 10^{-13}}$

$= 31.25 \times 10^{12}$

27. 00040.00

Sol. For P : $\dfrac{1}{-x} - \dfrac{1}{-30} = \dfrac{1}{f}$ …(1)

For Q : $\dfrac{1}{x} - \dfrac{1}{-60} = \dfrac{1}{f}$ …(2)

(2) + (1); gives

$\dfrac{1}{60} + \dfrac{1}{30} = \dfrac{2}{f}$

f = 40 cm

28. 00016.00

Sol. P.D. across capacitor is 16 V

$\therefore$ charge = 16 μC.

29. 00018.90

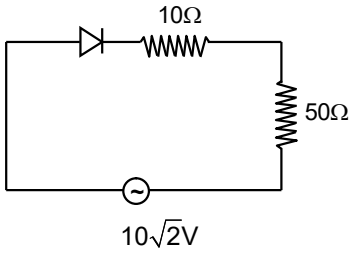
Range (18.90 to 18.93)

Sol. $I = \dfrac{20 - 0.7}{510} = 0.0378\,A$

Peak voltage across load

= 0.378 × 500

= 18.9 V



30. 00024.00

Sol. $m = \dfrac{f_0}{f_e} = \dfrac{\theta_i}{\theta_0}$

$\Rightarrow \theta_i = \dfrac{f_0}{f_e}.\theta_0$

$\theta_i = 24°$

Chemistry

PART – B

SECTION – A

31. C

Sol. $\psi = A\sin\left(\frac{2\pi x}{\lambda}\right)$

$\frac{d\psi}{dx} = \frac{2\pi}{\lambda}A\cos\left(\frac{2\pi x}{\lambda}\right)$

$\frac{d^2\psi}{dx^2} = -\left(\frac{2\pi}{\lambda}\right)A\sin\frac{2\pi x}{\lambda}$

$\frac{d^2\psi}{dx^2} = -\frac{4\pi^2\psi}{\lambda}$

$\therefore \frac{d^2\psi}{dx^2} + \frac{4\pi^2\psi}{\lambda^2} = 0$

32. B

33. D

Sol. Discharging reaction of lead storage battery.

$PbO_2(s) + Pb(s) + 4H^+ + 2SO_4^{2-} \longrightarrow 2PbSO_4(s) + 2H_2O(\ell)$

34. A

Sol. $\underset{\downarrow sp^3}{CH_4} + 2O_2 \longrightarrow \underset{\downarrow sp}{CO_2} + 2H_2O$

35. A

36. D

37. C

38. D

39. A

40. A

41. D

42. B

43. C

Sol. $\frac{225+375}{400}=\frac{600}{400}=1.50$

44. A

45. C

46. A

Sol. $2FeCl_3+3Ba(OH)_2\longrightarrow 2Fe(OH)_3\downarrow +3BaCl_2$

Brown colour

47. A

48. D

49. C

50. C

SECTION – B

51. 00003.05

52. 00000.04

Sol. $2CuSO_4+KI\longrightarrow K_2SO_4+Cu_2I_2+I_2$

$2Na_2S_2O_3+I_2\longrightarrow 2NaI+Na_2S_4O_6$

Milli eq. of $Na_2S_2O_3=$ Meq. of I_2 liberated $=$ Meq.of $CuSO_4=1$

$M_{hypo}\times 25=1\Rightarrow M_{hypo}=\frac{1}{25}=0.04$

53. 00002.44

Sol. $NiO(s)+H_2(g)\rightleftharpoons Ni(s)+H_2O(g)$

$K_p=\frac{P_{H_2O}}{P_{H_2}}$

Thus, we require value of $\log K_p$

$\Delta G^\circ=-2.303RT\log K_p$

$\log K_p=\frac{-\Delta G^\circ}{2.303RT}$

$=\frac{-(-8992)\,cal}{2.303\times 2\times 800}=2.44$

54. 00091.80

Sol. For isotonic solution, osmotic pressure are same as at same temperature.

$\pi_{glucose}=\pi_{glycerene}$

$C_{glucose}RT = C_{glycerene}RT$

$$\frac{n_{glucose}}{\text{vol. of solution}} = \frac{n_{glycerene}}{\text{vol. of solution}}$$

$\therefore$ Mol. Wt. of glycerene = 91.80 gm.

55. 00000.84

Sol. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ Vapour density $= 25$

1 0

$1-x$ $2x$

Vapour density = 25

$\therefore$ Mol. wt. $= 2 \times 25 = 50 = M'$

M = Molar mass before dissociation = 92

$$\alpha = \frac{M - M'}{(n-1)M'} = \frac{92-50}{1 \times 50} = 0.84$$

56. 00000.75

Sol. Let total volume is v ml

Activity = 0.240 μCi

After dispersion

$$\text{Activity per ml} = \frac{0.240\ \mu Ci/ml}{V}$$

$$\therefore \quad 3.2 \times 10^{-4}\ \mu Ci/ml = \frac{0.240\,\mu Ci/ml}{V}$$

$$V = \frac{0.240}{3.2 \times 10^{-4}} = \frac{0.240 \times 10^4}{3.2}$$

$v = 0.075 \times 10^4 = 750$ ml

$\therefore v = 0.75$ litre

57. 00006.00

58. 00000.10

Sol. $t_{1/2} = \frac{a}{2K}$ **= 0.1 sec**

(a = initial concentration)

59. 00004.00

60. 00021.90

Sol. Let the metal chloride MCl_n

$MCl_n + AgNO_3 \longrightarrow AgCl + M(NO_3)_n$

0.2 0.5 g

Meq. of MCl_n = Meq. of AgCl

$$\frac{0.2}{\text{Eq. wt. } MCl_n} \times 1000 = \frac{0.5}{\text{Eq.wt. of AgCl}} \times 1000$$

$$\text{Eq.wt.} MCl_n = \frac{0.2 \times 143.5}{0.5} = \frac{28.7}{0.5}$$

= 57.4

Eq. wt. of M + Eq. wt. of Cl^- = 57.4

Eq. wt. of M = 57.4 – 35.5 = 21.9 gm

Mathematics

PART – C

SECTION – A

61. C

Sol. $kx - 5y + 6 = 0$ should not be parallel to $3x - y + 2 = 0$ and $x + y + 1 = 0$ and it should also not pass through $\left(-\frac{1}{2}, \frac{1}{2}\right)$

62. D

Sol. $T_r = \frac{(2r+1)}{\frac{r^2(r+1)^2}{4}} = 4\left(\frac{(r+1)^2 - r^2}{r^2(r+1)^2}\right) = 4\left(\frac{1}{r^2} - \frac{1}{(r+1)^2}\right)$

$S_{10} = T_1 + T_2 + T_3 + ... T_{10} = 4\left(1 - \frac{1}{2^2} + \frac{1}{2^2} - \frac{1}{3^2} + ... \frac{1}{10^2} - \frac{1}{11^2}\right) = 4\left(1 - \frac{1}{121}\right) = \frac{480}{121}$

63. B

Sol. ${}^{23}C_1 + {}^{23}C_3 + {}^{23}C_5 + ... {}^{23}C_{23} = 2^{22}$

64. C

Sol. $\cos\frac{2\pi}{10} + \cos\frac{4\pi}{10} + \cos\frac{6\pi}{10} + ... \cos\frac{18\pi}{10}$

= Re$(1 + \alpha + \alpha^2 + ... \alpha^9)$ where α is the 10th root of unity

i.e., $\alpha = e^{\frac{i2\pi}{10}}$ and Re$(1 + \alpha + \alpha^2 + ... \alpha^9) = 0$

$\Rightarrow 1 + \cos\frac{2\pi}{10} + \cos\frac{4\pi}{10} + ... \cos\frac{18\pi}{10} = 0$

65. B

Sol. Equation of chord of contact $-x + ky = 4$

$\Rightarrow y = \frac{1}{k}x + \frac{4}{k}$. It touches the parabola $y^2 = 36$

$\therefore \frac{4}{k} = \frac{9}{1/k} \Rightarrow k^2 = \frac{4}{9} \Rightarrow k = \pm\frac{2}{3}$

66. C

Sol. $y = mx - 4m - 2m^3$ passes through $(k, 0)$

$\Rightarrow km - 4m - 2m^3 = 0 \Rightarrow 2m^3 + m(k - 4) = 0$

67. A

Sol. $\tan 2x + \tan 4x = \sqrt{3}(1 - \tan 2x \tan 4x)$

$\Rightarrow \frac{\tan 2x + \tan 4x}{1 - \tan 2x \tan 4x} = \sqrt{3} \Rightarrow \tan 6x = \sqrt{3}$

$\Rightarrow 6x = n\pi + \frac{\pi}{3} \Rightarrow x = \frac{n\pi}{6} + \frac{\pi}{18}$

$\Rightarrow x = 0, 1, 2, 3, 4, 5 \therefore$ 6 solutions

68. A

Sol. $\vec{a}\cdot\vec{c} = 3|\vec{c}| \Rightarrow |\vec{a}||\vec{c}|\cos\theta = 3|\vec{c}|$

$\therefore$ $\vec{c}$ is parallel to $\vec{a}$

Let $\vec{c} = \lambda\vec{a}$

$\therefore \left|\vec{b}\times\vec{c}\right| = \sqrt{212}$

$\Rightarrow \left|\vec{b}\times\lambda\vec{a}\right| = \sqrt{212}$

$\Rightarrow |\lambda|\left|\vec{b}\times\vec{a}\right| = \sqrt{212}$

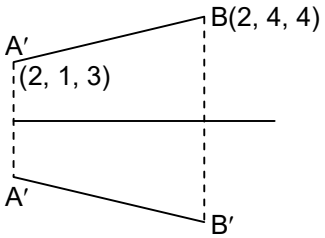
$\Rightarrow |\lambda|\left|-4\hat{i}+\hat{j}-6\hat{k}\right| = \sqrt{212}$

$\Rightarrow |\lambda|\times\sqrt{53} = \sqrt{212} \Rightarrow |\lambda| = 2 \Rightarrow \lambda = \pm 2$

$\therefore \vec{c} = 4\hat{i}+4\hat{j}-2\hat{k}$ or $\vec{c} = -4\hat{i}-4\hat{j}+2\hat{k}$

69. B

Sol. Let A′ and B′ be the image of A and B in the plane
$2x + y - 3z - 10 = 0$
On solving, we get
A′(6, 3, –3)
and B′(6, 6, –2)



70. C

Sol. $2\cos^2\theta - 5\sin\theta + 4\sin^2\theta$

$\Rightarrow 2\sin^2\theta - 5\sin\theta + 2 = 0$

$\Rightarrow \sin\theta = \frac{1}{2}$, 2 (rejected)

$\Rightarrow \theta_1 = \frac{\pi}{6}$ and $\theta_2 = \frac{5\pi}{6}$

71. A

Sol. ${}^{12}C_7 + {}^{11}C_6 + {}^{10}C_5 + {}^{9}C_4 + {}^{8}C_3 + {}^{7}C_2 + {}^{6}C_1 + 1$

$= {}^{12}C_7 + {}^{11}C_6 + {}^{10}C_5 + {}^{9}C_4 + {}^{8}C_3 + {}^{7}C_2 + {}^{6}C_1 + {}^{6}C_0$

Using ${}^{n}C_r + {}^{n}C_{r+1} = {}^{n+1}C_{r+1}$, we get the sum equal to ${}^{13}C_7$

72. D

Sol. $\frac{dx}{dy} + 2x = y^2e^{-y}$

$I.F = e^{\int 2dy} = e^{2y}$

$\therefore$ Solution is $xe^{2y} = \int y^2e^{-y}\times e^{2y}dy$

$\Rightarrow xe^{2y} = \int y^2e^{y}dy = \left(y^2-2y+2\right)e^{y} + c$

at x = 0, y = 1 $\Rightarrow$ c = –e

$\therefore xe^{2y} = \left(y^2-2y+2\right)e^{y} - e$

at y = 0, x = 2 – e

73. B

Sol. Probability that at most 3 machines are out of service is

$${}^7C_0\left(\frac{2}{3}\right)^7 + {}^7C_1\left(\frac{2}{3}\right)^6\left(\frac{1}{3}\right) + {}^7C_2\left(\frac{2}{3}\right)^5\left(\frac{1}{3}\right)^2 + {}^7C_3\left(\frac{2}{3}\right)^4\left(\frac{1}{3}\right)^3$$

$$\Rightarrow \frac{2^7 + 7\times 2^6 + 21\times 2^5 + 35\times 2^4}{3^7} = \frac{16k}{3^7} \Rightarrow k = 113$$

74. C

Sol. If each observation is multiplied by k, then mean also gets multiplied by k

If each observation is reduced by m, then mean also gets reduced by m

$\therefore$ 30k – m = 10

If each observation is multiplied by k, then standard deviation gets multiplied by |k|

If each observation is reduced by m, then standard deviation is not affected

$\therefore$ 6|k| = 2 $\Rightarrow k = \pm\frac{1}{3}$ and $\because$ $m \neq 0$ $\therefore$ m = –20

75. D

Sol. Let point p be (x_1, y_1). Thus equation of normal at (x_1, y_1) is $\frac{4x}{x_1} - \frac{9y}{y_1} = -5$

It cuts y axis at $\left(0, \frac{5y_1}{9}\right) \Rightarrow y_1 = 1$

Putting it in equation of ellipse, we get $9x_1^2 + 4\times 1^2 = 36$

$$\Rightarrow x_1 = \frac{4\sqrt{2}}{3}$$

76. C

Sol. Divide Num and Den by x^{21}

$$\int \frac{15x^{-6} + 7x^{-8}}{\left(1 + 3x^{-5} + x^{-7}\right)^3} dx$$

Let $1 + 3x^{-5} + x^{-7} = t \Rightarrow \frac{dt}{dx} = -15x^{-6} - 7x^{-8}$

$$\int \frac{-1}{t^3} dt = \frac{1}{2\left(1 + 3x^{-5} + x^{-7}\right)^2}$$

77. D

Sol. Centre of hyperbola is (0, 2) and one-one end of major axis is (3, 2)

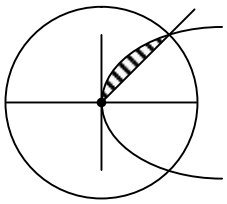
$\therefore$ a = 3 and $a^2 - b^2 = 4$ (radius of director circle)

$\Rightarrow b^2 = 5$

$$\therefore e = \frac{\sqrt{14}}{3}$$

78. D

Sol. Area of shaded region = $\int_0^2 \left(2\sqrt{2}x - 2x\right)dx = 2\sqrt{2}\frac{x^{\frac{3}{2}}}{\frac{3}{2}} - x^2\Big|_0^2 = \frac{4}{3}$



$\therefore$ Required area = $20\pi - \frac{4}{3}$

79. A

Sol. $\because$ It is $\frac{0}{0}$ form $\therefore$ using L-Hospital Rule

$$\lim_{x\to 1}\frac{\frac{\cos(x^2)}{2\sqrt{x}} - \cos\left(\frac{1}{x^4}\right)\times\left(\frac{-1}{x^2}\right)}{2x-4} = \frac{\frac{\cos 1}{2} - \cos(1)(-1)}{-2} = \frac{-3\cos 1}{4}$$

80. B

Sol. α is a root of $f'(x) = 0$ also $3x^2 + 8x - 3 = 0 \Rightarrow x = -3, \frac{1}{3}$

$\therefore \alpha$ can be -3 or $\frac{1}{3}$ $\therefore f(-3) = 0 \Rightarrow k = -18$ and $f\left(\frac{1}{3}\right) = 0$

$\Rightarrow k = \frac{14}{27}$

SECTION – B

81. 00001.58

Sol. $$2\lim_{n\to\infty} L = 2\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}\frac{1}{\left(1+\frac{r}{n}\right)\left(1+\frac{2r}{n}\right)\left(1+\frac{3r}{n}\right)}$$

$$= 2\int_0^1 \frac{1}{(1+x)(1+2x)(1+3x)}dx = 2\int_0^1 \frac{1/2}{(1+x)} - \frac{4}{(1+2x)} + \frac{9/2}{(1+3x)}dx$$

$$= 2\left(\frac{1}{2}\ln(1+x) - 2\ln(1+2x) + \frac{3}{2}\ln(1+3x)\right)\Big|_0^1 = \ln 2 - 4\ln 3 + 3\ln 4 = \ln\frac{128}{81}$$

$\therefore e^{\lim_{n\to\infty} 2L} = \frac{128}{81} = 00001.58$

82. 00650.00

Sol. $\sum_{k=1}^{12} k^2 = \frac{12\times 13\times 25}{6} = 650$

83. 00003.00

Sol. $f(1) = f(3) \Rightarrow \frac{1+k}{3} = \frac{9+k}{9} \Rightarrow k = 3$

$\therefore f'(x) = \frac{3x}{x^2+3}\left(\frac{1}{3} - \frac{1}{x^2}\right) = 0 \Rightarrow x = \sqrt{3}$

84. 00002.00

Sol. $\left(\cos^{-1}x + \cos^{-1}y\right)^2 = 4\pi^2 \quad \Rightarrow \cos^{-1}x + \cos^{-1}y = 2\pi$

85. 00000.27

Sol. $C_1(2, -1)$, $C_2(6, 2)$, $\therefore C_1C_2 = 5$

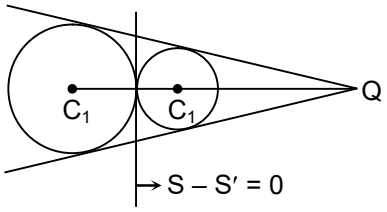
$r_1 = 3$, $r_2 = 2$

$\therefore C_1C_2 = r_1 + r_2$

$\therefore$ 3 common tangents

$S - S' = 0$

$\Rightarrow 4x + 3y - 10 = 0$



$\therefore$ One of the slopes is $-\frac{4}{3}$

Q divides C_1C_2 in the ratio $r_1 : r_2$ externally $\therefore$ Q is (14, 8)

Any line passing through (14, 8) is $y - 8 = m(x - 14)$

$\Rightarrow mx - y + 8 - 14m = 0$

For it to be tangent to the circle $\frac{|2m + 1 + 8 - 14m|}{\sqrt{m^2 + 1}} = 3$

$\Rightarrow 15m^2 - 24m + 8 = 0 \Rightarrow m_1 + m_2 = \frac{24}{15} = \frac{8}{5}$

$\therefore$ Sum of slopes is $\frac{8}{5} - \frac{4}{3} = \frac{4}{15}$

86. 00007.00

Sol. System has infinitely many solutions

$$\therefore D = 0 \Rightarrow \begin{vmatrix} 1 & 2 & k \\ 1 & 1 & 1 \\ 3 & 2 & 1 \end{vmatrix} = 0 \Rightarrow 1(1 - 2) - 2(1 - 3) + k(2 - 3) = 0 \therefore k = 3$$

$$Dx = 0 \Rightarrow \begin{vmatrix} m & 2 & 3 \\ 6 & 1 & 1 \\ 14 & 2 & 1 \end{vmatrix} = 0 \Rightarrow m(1 - 2) - 2(6 - 14) + 3(12 - 14) = 0$$

$-m + 16 - 6 = 0 \Rightarrow m = 10$

87. 00000.70

Sol. $\lim\limits_{x \to 0} \frac{f(x)}{x^3} = 4 \Rightarrow f(x) = 4x^3 + ax^4 + bx^5$

$f'(x) = 12x^2 + 4ax^3 + 5bx^4$

$f'(1) = 0 \Rightarrow 12 + 4a + 5b = 0$ and $f'(2) = 0 \Rightarrow 48 + 32a + 80b = 0$

On solving, we get $a = -\frac{9}{2}$ and $b = \frac{6}{5}$

$\therefore f(x) = \frac{6}{5}x^5 - \frac{9}{2}x^4 + 4x^3$

88. 00000.50

Sol. $b_{ij} = 2^{(i+j-1)}a_{ij}$

$$B = \begin{bmatrix} 2a_{11} & 2^2a_{12} & 2^3a_{13} \\ 2^2a_{21} & 2^3a_{22} & 2^4a_{23} \\ 2^3a_{31} & 2^4a_{32} & 2^5a_{33} \end{bmatrix}$$

$$\Rightarrow |B| = 2\times 2^2\times 2^3\begin{vmatrix} a_{11} & 2a_{12} & 2^2a_{13} \\ a_{21} & 2a_{22} & 2^2a_{23} \\ a_{31} & 2a_{32} & 2^2a_{33}\end{vmatrix} = 2\times 2^2\times 2^3\times 2\times 2^2\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33}\end{vmatrix} = 2^9|A|$$

89. 00626.00

Sol.

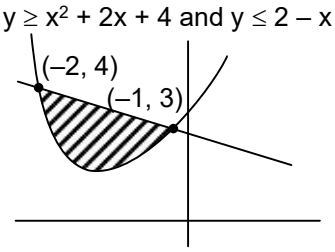
Category	Selection	Arrangement
all different	${}^6C_4$	${}^6C_4\times 4! = 360$
2 alike 2 alike	${}^2C_2$	${}^2C_2\times\frac{4!}{2!2!} = 6$
3 alike 1 different	${}^1C_1\times{}^5C_1$	$1\times 5\times\frac{4!}{3!} = 20$
2 alike 2 different	${}^2C_1\times{}^5C_2$	${}^2C_1\times{}^5C_2\times\frac{4!}{2!} = 240$
		Total = 626

90. 00000.17

Sol. $x^2 + 2x + 4 = 2 - x$

$x^2 + 3x + 2 = 0$

$x = -1, -2$



Required area $= \int_{-2}^{-1}(2-x)-(x^2+2x+4)dx$

$$= \int_{-2}^{-1}(-2-3x-x^2)dx = -2x-\frac{3x^2}{2}-\frac{x^3}{3}\Big|_{-2}^{-1}$$

$$= -\left(2x+\frac{3x^2}{2}+\frac{x^3}{3}\right)\Big|_{-2}^{-1} = -\left(\left(-2+\frac{3}{2}-\frac{1}{3}\right)-\left(-4+6-\frac{8}{3}\right)\right) = -\left(\frac{-5}{6}-\left(\frac{-2}{3}\right)\right) = \frac{1}{6}$$